
CISCO FTD

Clustering (Active/Active) FMC-Managed Configuration Guide

A complete lab walkthrough covering FTD Active/Active clustering — from the differences between HA and clustering, through VXLAN/VTEP network planning, cluster wizard configuration in FMC, deployment, CLI verification, and policy assignment for a distributed firewall architecture.

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Category: Network Security Lab · Portfolio Project

Platform: Cisco Firepower (FMC + FTD v7.3.1 · Active/Active Cluster)

1. Introduction

FTD Clustering enables two or more Firepower Threat Defense appliances to operate as a single logical firewall in an Active/Active configuration. Unlike High Availability (Active/Standby) where only one unit processes traffic, clustering allows all members to actively forward traffic simultaneously, providing both scalability and redundancy.

All cluster management — formation, policy deployment, and monitoring — is handled centrally through FMC. The cluster appears as a single device in FMC, and policies are applied cluster-wide.

► 1.1 Clustering vs. High Availability

Understanding the difference between these two architectures is critical for choosing the right deployment model:

Feature	HA (Active/Standby)	Cluster (Active/Active)
Number of FTDs	Exactly 2	2 or more
Traffic Processing	Only active unit forwards	All members forward simultaneously
Primary Use Case	Stateful failover / redundancy	Scalability & throughput aggregation
Connection Handling	Standby holds state only	Connections distributed across members
Scaling	No horizontal scaling	Add members to increase capacity
Managed By	FMC	FMC

Clustering is different from HA, but FMC manages both. Choose HA when you need simple redundancy; choose clustering when you need performance scaling.

2. Lab Environment

► 2.1 Network Topology

The lab uses two FTD appliances (FTD1 and FTD2) managed by a single FMC, with a dedicated cluster control link (G0/2), inside and outside data interfaces, and a testing client connected through a layer-2 switch.

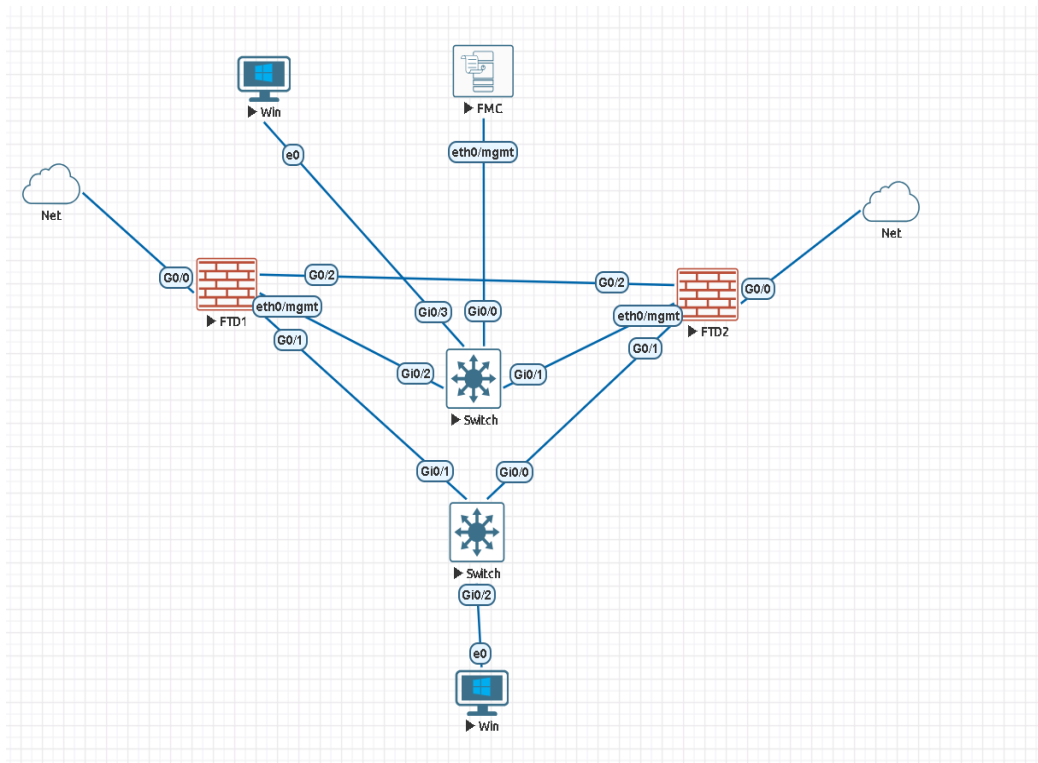


Figure 1 — Cluster Lab Network Topology

2.2 Registered Devices

Both FTDs are registered to FMC as individual devices before cluster creation. They must be identical in model (Firewall Threat Defense for KVM), version (7.3.1), and licensing (Essentials):

☐	Name	Model	Ver...	Chassis	Licenses	Access Control Policy	Auto RollBack
☐	Un grouped (2)						
☐	FTD1 Snort 3 192.168.10.2 - Routed	Firewall Threat Defense for...	7.3.1	N/A	Essentials	to_internet	🔄
☐	FTD2 Snort 3 192.168.10.3 - Routed	Firewall Threat Defense for...	7.3.1	N/A	Essentials	to_internet	🔄

Figure 2 — Both FTDs registered and healthy in FMC

3. Preparation

3.1 Step 1 — Prepare FTDs for Clustering

Before creating the cluster, ensure both FTDs have identical configurations:

- Security zones must match (Inside / Outside)
- IP addressing and subnets must be consistent
- Default routes and NAT rules must be identical
- Access control policies should match

Cluster Control Link

Prepare a dedicated cluster control link on GigabitEthernet0/2 with assigned IPs for each FTD:

Device	Interface	IP Address	Purpose
FTD1	G0/2	192.168.250.1/30	Cluster Control Link
FTD2	G0/2	192.168.250.2/30	Cluster Control Link

This link is used for cluster communication — connection sync, NAT table sync, VXLAN tunnels, and heartbeats. It is similar in purpose to the HA failover link, but designed for cluster control.

4. Create Cluster in FMC

► 4.1 Step 2 — Launch the Cluster Wizard

Navigate to Devices → Device Management → Add → Cluster:

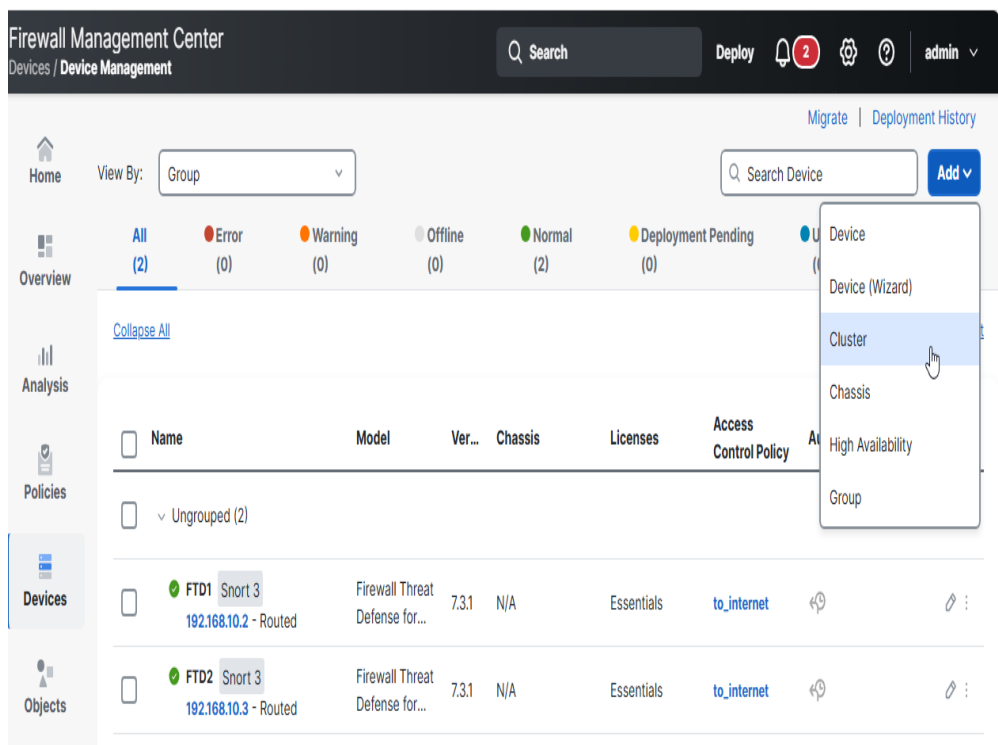


Figure 3 — FMC Device Management: Add → Cluster option

► 4.2 Step 3 — Configure Cluster Parameters

The Add Cluster Wizard requires several fields. Each one serves a specific purpose in building the VXLAN-based cluster overlay:

Add Cluster Wizard



1 Configuration — 2 Summary



Create a cluster for supported models. Note: For the Firepower 4100/9300 and threat defense virtual (AWS/GCP/Azure), use the Add Device option. Make sure connected switches match the MTUs for data interfaces and the cluster control link interface.

Cluster Name *

Cluster-Test

Cluster Key

.....

.....

Control Node

You can form the cluster with just the control node to reduce formation time.

Node *

FTD2

VXLAN Network Identifier (VNI) Network

10.10.10.0

/ 27 (30 addresses)

Virtual Tunnel Endpoint (VTEP) Network

10.10.20.0

/ 27 (30 addresses)

Cluster Control Link *

GigabitEthernet0/2

VTEP IPv4 Address *

10.10.20.1

Priority *

1

Data Nodes (Optional)

Data node hardware needs to match the control node hardware.

[Add a data node](#)

Activate Windows
Go to Settings to activate Windows.
Cancel Continue

Figure 4 — Add Cluster Wizard: Configuration page

Field 1: Cluster Name

The logical name of the cluster object inside FMC. Used for identification only — it does not affect the FTDs or the network.

- Example: Cluster-Test or FTD-Cluster-LAB

Field 2: Cluster Key

An encryption key used to secure and authenticate cluster communication between members. All members must share the same key.

- Prevents unauthorized nodes from joining the cluster
- Encrypts control-plane synchronization over the CCL
- Enter the same key twice (confirmation field)

Field 3: Control Node

Select which FTD becomes the cluster control node during initial formation. The control node manages cluster creation, holds the initial VXLAN VNI address assignments, and is the first member other nodes join.

- Must be online and reachable by FMC
- Other members join after the control node is established

Field 4: VXLAN Network Identifier (VNI) Network

An internal virtual network used inside the cluster for data synchronization via VXLAN tunnels. The cluster uses this overlay to sync NAT tables, connection states, routing information, and bucket distribution tables.

- Must be unique and not overlap with any real network
- Must not be routable in your production network

- Best practice: use reserved ranges like 10.10.x.x
- Example: 10.10.10.0/27

Field 5: Virtual Tunnel Endpoint (VTEP) Network

A separate virtual network used for VXLAN encapsulation endpoints. Each cluster node receives a VTEP address from this range.

- Must also be unique and non-overlapping
- Example: 10.10.20.0/27

Field 6: VTEP IPv4 Address

The VTEP address assigned to the control node. Must be the first usable IP in the VTEP network.

- If VTEP Network = 10.10.20.0/27, then VTEP IPv4 = 10.10.20.1
- The second member automatically receives 10.10.20.2

Field 7: Cluster Control Link (CCL)

The physical interface used for all cluster synchronization — connection tables, NAT tables, VXLAN tunnels, and heartbeats. This must be the dedicated direct link between your FTDs.

- In this lab: GigabitEthernet0/2
- The CCL uses a higher MTU than data interfaces (set to 1654)
- Ensure the connected switch ports have matching MTU

Field 8: Priority

Determines which node has preference to become the master/control member in the cluster.

- Lower number = higher priority
- Control node should have priority 1
- Additional nodes: 5, 10, etc.

► 4.3 Why FMC Needs All These Fields

To create a functional cluster, FMC builds several internal structures:

- VXLAN mapping for overlay communication
- Internal tunnel IPs for each member
- Virtual MAC calculation for seamless traffic handling
- Bucket distribution table for flow allocation
- Control-plane sync mappings between members

These structures depend on VNI (VXLAN overlay), VTEP (tunnel endpoints), CCL (physical transport), Priority (master selection), and the Cluster Key (encryption and authentication). All are mandatory for the cluster to operate as an Active/Active distributed firewall.

5. Deployment & Verification

► 5.1 Step 4 — Deploy Configuration

After completing the wizard, FMC presents a summary page confirming all cluster parameters, including the jumbo frame requirement for the cluster control link:

Add Cluster Wizard

1 Configuration — 2 Summary

⚠

Clustering requires jumbo frames for the cluster control link. If you did not enable jumbo frames at deployment or have not previously enabled jumbo frames by setting the MTU of an interface above 1500, you need to manually reboot each node after the cluster is formed and healthy. Use the 'show jumbo-frame reservation' command on the device to check jumbo frame status.

⚠

The cluster control link uses a higher MTU than data interfaces. The current max data MTU is 1500; the cluster control link MTU will be set to 1654. Make sure you set the switch MTU to match, otherwise cluster formation will fail. [More info](#)

Cluster Name: **Cluster-Test**

Cluster Key: (Specified)

Control Node

Name	Priority	VNI Network	VTEP IPv4 Address	Cluster Control Link	VTEP Network
FTD2	1	10.10.10.0/27	10.10.20.1	GigabitEthernet0/2	10.10.20.0/27

Figure 5 — Cluster Wizard Summary page (jumbo frame warnings)

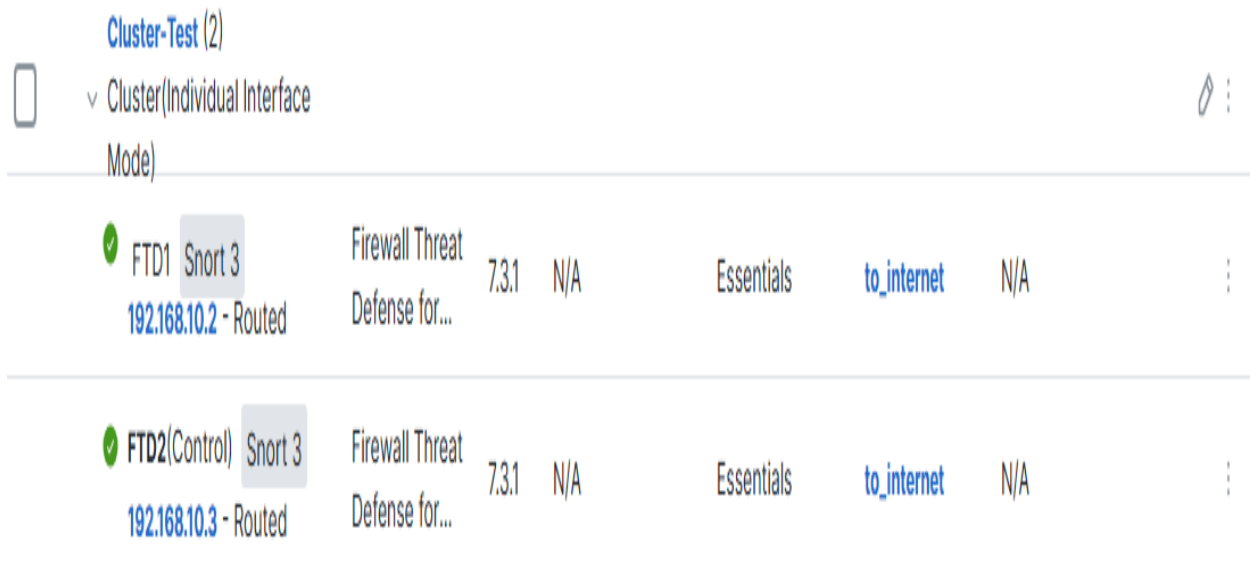
Click Deploy → Deploy Now. Both FTDs will reboot as part of cluster formation. After reboot, the cluster state will show:

- Member 1 → Active
- Member 2 → Active
- Traffic load balanced across members per the configured cluster algorithm

► 5.2 CLI Verification

Verify cluster status from either FTD CLI:

```
> show cluster info
```



Cluster-Test (2)									
Cluster (Individual Interface Mode)									
✓	FTD1	Snort 3	Firewall Threat Defense for...	7.3.1	N/A	Essentials	to_internet	N/A	⋮
		192.168.10.2 - Routed							
✓	FTD2(Control)	Snort 3	Firewall Threat Defense for...	7.3.1	N/A	Essentials	to_internet	N/A	⋮
		192.168.10.3 - Routed							

Figure 6 — show cluster info: FTD1 (DATA_NODE) and FTD2 (CONTROL_NODE)

Key observations from the CLI output:

- Cluster name: Cluster-Test, Interface mode: individual
- FTD1 is in state DATA_NODE (ID: 1, CCL IP: 10.10.10.2)
- FTD2 is in state CONTROL_NODE (ID: 0, CCL IP: 10.10.10.1)
- Both nodes: 4 cores / 8192 MB RAM, version 9.19(1)200

► 5.3 FMC Verification

In FMC, the cluster appears as a single logical device under Device Management:

```

Cluster Cluster-Test: On
  Interface mode: individual
Cluster Member Limit : 16
  This is "FTD1" in state DATA_NODE
    ID      : 1
    Version  : 9.19(1)200
    Serial No.: 9AAFMG286F5
    CCL IP   : 10.10.10.2
    CCL MAC  : 5000.0002.0004
    Module   : NGFWv
    Resource  : 4 cores / 8192 MB RAM
    Last join : 19:19:09 UTC Nov 25 2025
    Last leave: N/A
Other members in the cluster:
  Unit "FTD2" in state CONTROL_NODE
    ID      : 0
    Version  : 9.19(1)200
    Serial No.: 9AUQHL9LQFG
    CCL IP   : 10.10.10.1
    CCL MAC  : 5000.0003.0004
    Module   : NGFWv
    Resource  : 4 cores / 8192 MB RAM
    Last join : 19:15:17 UTC Nov 25 2025
    Last leave: N/A

```

Figure 7 — Cluster-Test in FMC: FTD1 (Data) + FTD2 (Control)

Verify from FMC → Device Management → Cluster → Status:

- Both members online
- Cluster Control Link green
- Sync Status: Completed

6. Configure Policies for Cluster

Once the cluster is formed and verified, assign policies to the cluster object (not individual FTDs):

1. Access Control Policies (ACP) → assign to the cluster
2. NAT Rules → assign to the cluster
3. Routing → ensure cluster interface IPs are reachable

Once deployed, the cluster acts as a single logical firewall. All policies are applied centrally through FMC and distributed to every member automatically.

7. Why Clustering?

► 7.1 Performance & Scalability

- Cluster distributes flows across multiple FTD nodes
- Increases overall throughput, concurrent sessions, and connections-per-second (CPS)
- Provides horizontal scaling — add more appliances as demand grows

► 7.2 High Availability

- If a node fails, remaining nodes continue handling existing and new traffic
- Superior to classic HA which only uses one appliance at a time
- No single point of failure in the data plane

► 7.3 Single Logical Device

- FMC manages the entire cluster as one firewall
- NAT, Access Control, Routing, VPN, and all policies are applied cluster-wide
- Simplifies operational management compared to managing individual devices

8. Summary

Component	Details
Cluster Mode	Active/Active (all members forward traffic)
Managed By	Firepower Management Center (FMC)
Overlay Technology	VXLAN (VNI + VTEP networks)
Control Link	GigabitEthernet0/2 (dedicated CCL)
Authentication	Shared Cluster Key (encryption)
Node Roles	Control Node (master) + Data Nodes
Priority	Lower number = higher priority for control
MTU Requirement	CCL set to 1654; switch ports must match
Verification	show cluster info (CLI) / FMC Device Management
Policy Assignment	Applied to cluster object, distributed to all members
Typical Use Case	High-throughput environments requiring horizontal scaling